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Operating System

INTER-PROCESS COMMUNICATION USING SHARED MEMORY

Aim:
To write a C program to do Inter Process Communication (IPC) using shared memory between sender process and receiver process.

Algorithm:

sender
1. Set the size of the shared memory segment
2. Allocate the shared memory segment using shmget
3. Attach the shared memory segment using shmat
4. Write a string to the shared memory segment using sprintf
5. Set delay using sleep
6. Detach shared memory segment using shmdt

receiver
1. Set the size of the shared memory segment
2. Allocate the shared memory segment using shmget
3. Attach the shared memory segment using shmat
4. Print the shared memory contents sent by the sender process.
5. Detach shared memory segment using shmdt

Program Code:

Sender.c

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <unistd.h>

int main() {
    int shmid;
    char *str;

    key_t key = 1234;

    shmid = shmget(key, 1024, 0666 | IPC_CREAT);
```

```
    str = (char*) shmat(shmid, NULL, 0);

    sprintf(str, "Welcome to Shared Memory");

    printf("Message sent!\n");

    sleep(10);

    shmdt(str);

    return 0;
}
```

Receiver.c

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
int main() {
    int shmid;
    char *str;
    key_t key = 1234;

    shmid = shmget(key, 1024, 0666);

    str = (char*) shmat(shmid, NULL, 0);

    printf("Message Received: %s\n", str);

    shmdt(str);

    return 0;
}
```

Sample Output
Terminal 1
[root@localhost student]# gcc sender.c -o sender
[root@localhost student]# ./sender

Terminal 2
[root@localhost student]# gcc receiver.c -o receiver
[root@localhost student]# ./receiver
Message Received: Welcome to Shared Memory
[root@localhost student]#

Result:

gcc sender.c -o sender
./sender
Message sent!

gcc receiver.c -o receiver
./receiver
Message Received: Welcome to Shared Memory